🔎 ProfitProbe

AI business analyst for a Myntra-style fashion e-commerce company: every department says it is on target, yet profit has fallen three quarters in a row. (Synthetic illustrative data, 24 months to Sep 2026.)

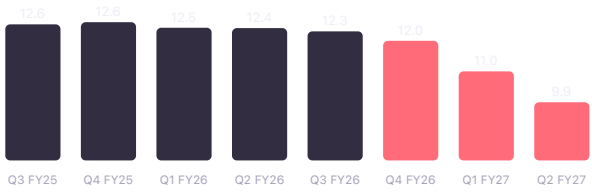
Margin now
9.9%

Fall vs pre-decline
-2.4 pts

Quarters of material decline
3

Explained by the chain
61%

Contribution margin by quarter (%)



What each department reports

Dept / KPI	Q3 FY26	Q4 FY26	Q1 FY27	Q2 FY
Marketing Orders vs plan ≥100%	Met 100	Met 103	Met 113	M 118
HR Headcount fill ≥95%	Met 96	Met 96	Met 96	M 96
Operations On-time delivery ≥94%	Met 94	Met 95	Met 95	M 95
Finance Revenue vs plan ≥100%	Met 100	Met 103	Met 113	M 118

Hidden cross-functional chain behind the decline



No department owns this loop: it starts in Marketing, passes through HR and Operations, and shows up in Finance. The decline began around May 2026.

KPI evidence vs margin change

KPI	Dept	Ra
Return rate %	Operations	-0.
Conversion %	Marketing	-0.
Pick error %	Operations	-0.
Attrition %	HR	-0.
Revenue ₹cr	Finance	-0.
Orders (lakh)	Marketing	-0.
Discount depth %	Marketing	-0.
On-time delivery %	Operations	-0.
Gross margin %	Finance	0.9
Overtime h/head	HR	-0.
Working capital days	Finance	-0.
Training spend ₹cr	HR	-0.
Cost per order ₹	Operations	-0.
CAC ₹	Marketing	0.1

What-if: break the chain

Fix at: Discount depth %

Reverse **50**% of its rise since the decline began

+0.74 pts margin recovery

Chain-mediated estimate from Discount depth % rising 6.4 units since onset. Total fall to recover: 2.4 pts; the rest comes from direct discount cost.

Headcount fill %	HR	-0.

Analyst memo (LLM)

Generate CFO memo

The LLM sees only the computed evidence.

Click to generate. Without LLM access the page shows a rule-based summary.

Ask e.g. Why is HR linked to the profit drop?

Ask

Answers are grounded in the evidence.

Algorithm used by the analyst

1. **Integrate** 15 monthly KPIs from Marketing, HR, Operations and Finance with the profit outcome (contribution margin %).
2. **Paradox test**: score each department against its own targets (all show as met) and detect the profit decline: trailing quarters with a drop of more than 0.2 pts.
3. **Change-point**: split the margin series at the point minimizing within-segment squared error to date the onset.
4. **Detrend**: use month-over-month changes, because most KPIs trend and raw correlations are misleading.
5. **Drivers**: correlate each KPI change with margin change at lags 0–4; significant if `|r| > 2/√n` .
6. **Cross-functional graph**: draw an edge A→B when A leads B by one month, `|r| > 0.7` and A→B beats B→A by 0.15 (direction test). Revenue is excluded because it duplicates orders.
7. **Root-cause chain**: search all simple paths and keep the one with the strongest total link strength, ending at a KPI that drives margin.
8. **Quantify**: estimate each link's slope by OLS and multiply along the chain to get the margin loss mediated by the chain; simulate fixes at any link.
9. **Narrate**: the LLM receives the evidence pack and writes a memo that states the chain, the share explained and a validation step.